

Experiment 1 — User Decision Prediction (NON-DS BLOCKS)

NON-DS blocks only (alert_system == 0) N=62,000

Calibration: isotonic Cal ratio per-user: 10%

Ablation Summary (test set)

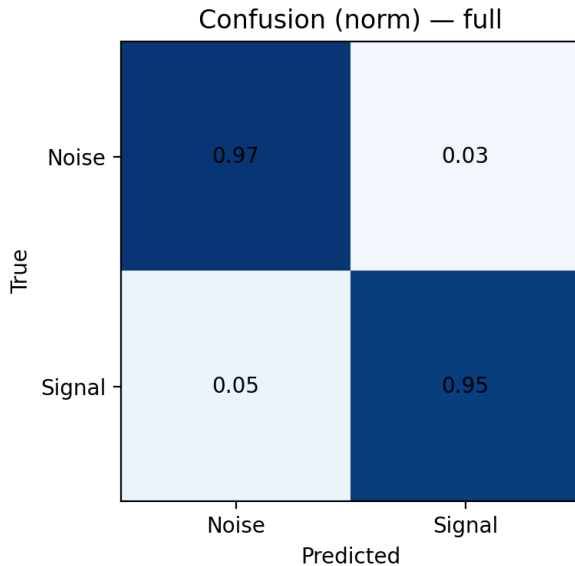
catboost_full | Acc@0.5=0.9648 F1@0.5=0.9494 AUC=0.9928 ECE=0.0096 t*=0.43 Acc@t*=0.9647 F1@t*=0.9493

full | Acc@0.5=0.9641 F1@0.5=0.9483 AUC=0.9932 ECE=0.0088 t*=0.43 Acc@t*=0.9601 F1@t*=0.9434

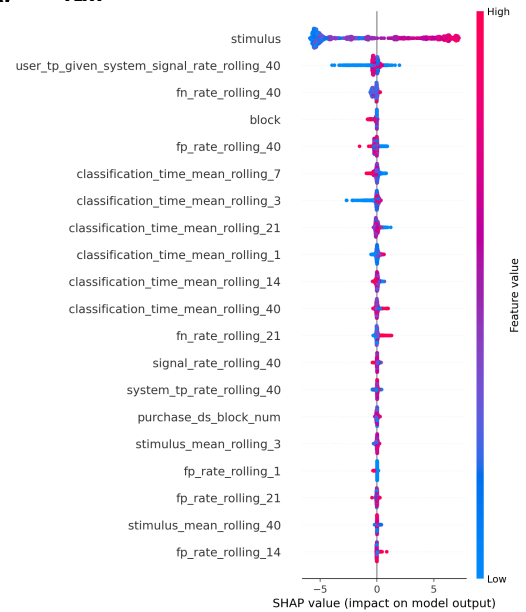
no_historical | Acc@0.5=0.9541 F1@0.5=0.9354 AUC=0.9766 ECE=0.0267 t*=0.30 Acc@t*=0.9531 F1@t*=0.9341

no_stimulus | Acc@0.5=0.9360 F1@0.5=0.9033 AUC=0.9827 ECE=0.0443 t*=0.36 Acc@t*=0.9394 F1@t*=0.9098

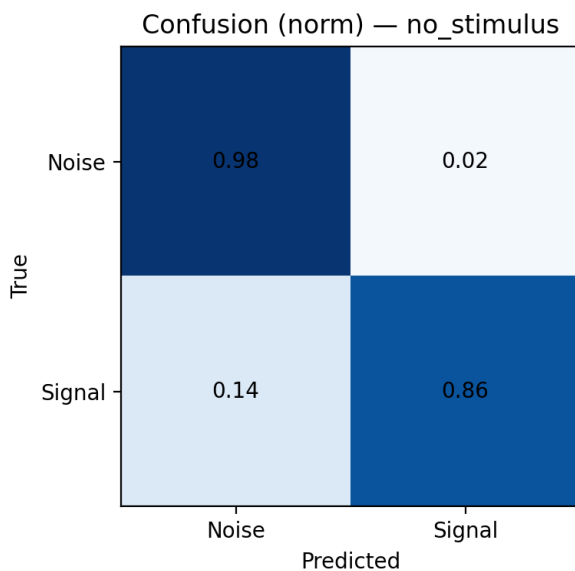
Confusion (norm) — full



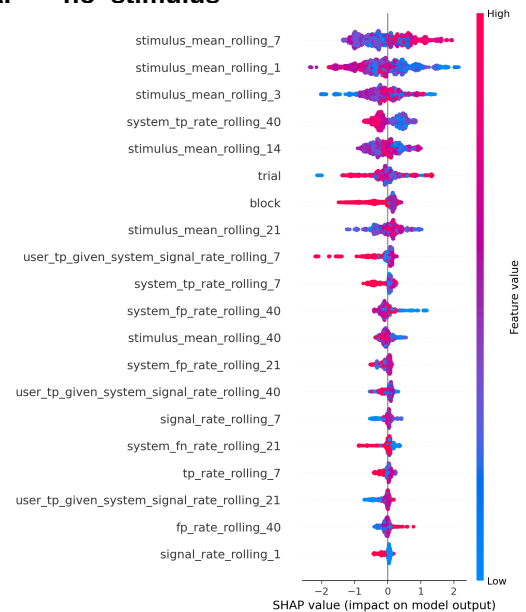
SHAP — full



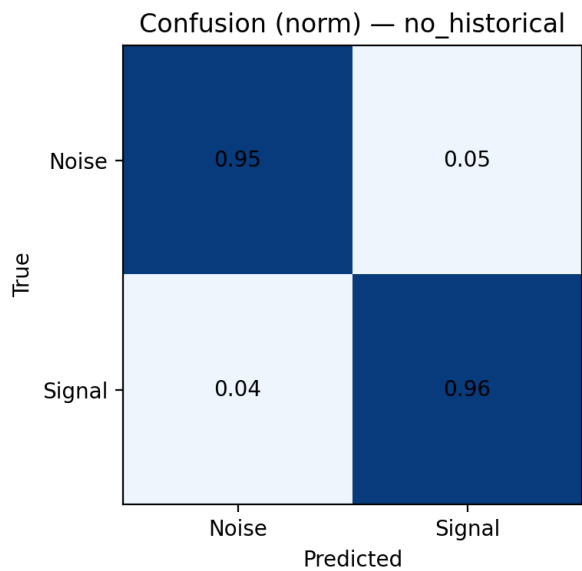
Confusion (norm) — no stimulus



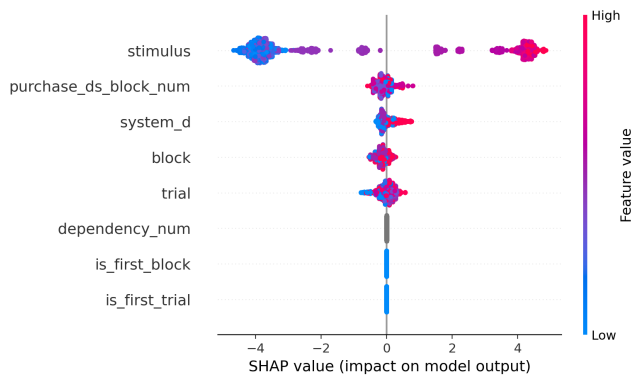
SHAP — no stimulus



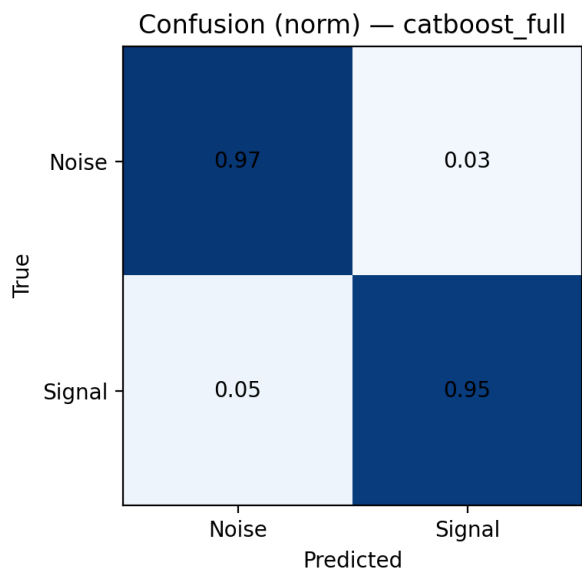
Confusion (norm) — no_historical



SHAP — no_historical



Confusion (norm) — catboost full



SHAP — catboost full

